

Input data

Features: age, gender, bmi, smoking_status, alcohol_consumption, exercise_level, diet_type, sun_exposure, income_level, latitude_region, vitamin_a_percent_rda, vitamin_c_percent_rda, vitamin_d_percent_rda, vitamin_e_percent_rda, vitamin_b12_percent_rda, folate_percent_rda, calcium_percent_rda, iron_percent_rda, hemoglobin_g_dl, serum_vitamin_d_ng_ml, serum_vitamin_b12_pg_ml, serum_folate_ng_ml, symptoms_count, has_night_blindness, has_fatigue, has_bleeding_gums, has_bone_pain, has_muscle_weakness, has_numbness_tingling, has_memory_problems, has_pale_skin, disease_diagnosis, has_multiple_deficiencies (total: 33 features)

Meta attributes: symptoms_list

Output data

Features: vitamin_a_percent_rda, hemoglobin_g_dl, vitamin_c_percent_rda, vitamin_e_percent_rda, vitamin_b12_percent_rda, vitamin_d_percent_rda, folate_percent_rda, calcium_percent_rda, iron_percent_rda, serum_vitamin_d_ng_ml, serum_vitamin_b12_pg_ml, serum_folate_ng_ml, has_muscle_weakness, has_night_blindness, has_bleeding_gums, has_fatigue, has_bone_pain, has_numbness_tingling, has_memory_problems, has_pale_skin, diet_type (total: 21 features)

Meta attributes: symptoms_list

Target: disease_diagnosis

Removed: 11 (exercise_level, has_multiple_deficiencies, sun_exposure, gender, age, bmi, alcohol_consumption, latitude_region, smoking_status, income_level, symptoms_count)

On va s'intéresse seulement parametre biologique et nutritionnel

Name: Random Forest

Model parameters

Number of trees: 100

Maximal number of considered features: unlimited

Replicable training: No

Maximal tree depth: unlimited

Stop splitting nodes with maximum instances: 5

python -> sklearn : parlons aux prof pour algo !!

Confusion matrix for Random Forest (showing proportion of actual)

		Predicted					Σ
		Anemia	Healthy	Night_Blindness	Rickets_Osteomalacia	Scurvy	
Actual	Anemia	92.5 %	0.0 %	0.0 %	7.5 %	0.0 %	1245
	Healthy	0.3 %	96.2 %	0.0 %	3.6 %	0.0 %	1509
	Night_Blindness	10.7 %	1.6 %	80.3 %	7.4 %	0.0 %	122
	Rickets_Osteomalacia	10.3 %	1.6 %	0.4 %	87.7 %	0.1 %	1029
	Scurvy	11.6 %	7.4 %	4.2 %	6.3 %	70.5 %	95
Σ		1286	1476	106	1064	68	4000

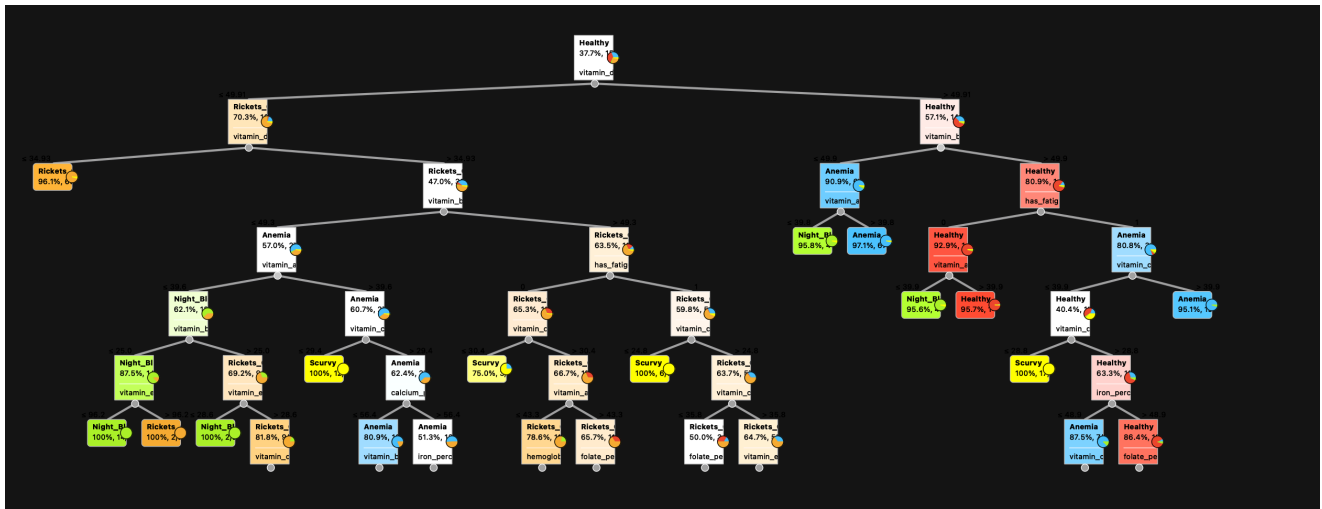
Night blindees et scurvy n'ont pas beaucoup des patients, on va cherche solutions pour optimisations

Healty, Anemia et Rickets sont class success

Tree size: 281 nodes, 141 leaves

Edge widths: Fixed

Target class: None



Anemie -> hemoglobin + fatigue + b12
 Rickets -> vitamin D !!!!!!! IMPORTANT
 Scurtvy -> vitamin C
 Night blindes -> vitamin A